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Contents

Question 1

Let's create a file and check it

```
[root@rhel-9 ~ #] touch /root/foo.sh
[root@rhel-9 ~ #] ls
foo.sh  MY FILE  qc  xx dsd
```

Then we give execute permission to this file:

```
[root@rhel-9 ~ #] chmod +x foo.sh
[root@rhel-9 ~ #]
```

In the script file:

```
#!/bin/bash

if [[ $1 == "RedHat" ]] ; then
        echo "fedora"
elif [[ $1 == "fedora" ]] ; then
        echo "redhat"
elif [[ -z $1 ]] ; then
        echo "Please enter argument" >&2
else
        echo "/root/foo.sh redhat|fedora"
fi
```

BTW, here '>&2' means redirecting the result to stderr.

When we check the result:

```
[root@rhel-9 ~ #] ./foo.sh RedHat
fedora
[root@rhel-9 ~ #] ./foo.sh fedora
redhat
[root@rhel-9 ~ #] ./foo.sh ubuntu
/root/foo.sh redhat|fedora
[root@rhel-9 ~ #] ./foo.sh
Please enter argument
[root@rhel-9 ~ #]
```

Question 2

Let’s create a file and give execute permission to it:

```
[root@rhel-9 ~ #] touch nums.sh
[root@rhel-9 ~ #] chmod +x nums.sh
[root@rhel-9 ~ #] ll | grep "nums.sh"
-rwxr-xr-x. 1 root root    0 May  7 03:50 nums.sh
[root@rhel-9 ~ #]
```

The script file:

```
#!/bin/bash

echo -n "Enter number: ";
read number;

if [[ $number -gt 200 ]] ; then
        echo "more than 200"
elif [[ $number -lt 100 ]] ; then
        echo "less than 100"
elif [[ $number == 100 || $number == 200 ]] ; then
        echo "number is $number"
else
        echo "bingo"
fi
```

BTW, there is no condition about if the number is equal to 100 or 200, so I have added it myself for fun.

```
[root@rhel-9 ~ #] ./nums.sh
Enter number: 250
more than 200
[root@rhel-9 ~ #] ./nums.sh
Enter number: 80
less than 100
[root@rhel-9 ~ #] ./nums.sh
Enter number: 120
bingo
[root@rhel-9 ~ #] ./nums.sh
Enter number: 100
number is 100
[root@rhel-9 ~ #] ./nums.sh
Enter number: 200
number is 200
```

Question 3

Let's create a file and give an execute permission to it:

```
[root@rhel-9 /script #] touch script.sh
[root@rhel-9 /script #] chmod +x script.sh
[root@rhel-9 /script #] ll | grep "script.sh"
-rwxr-xr-x. 1 root root  20 May  7 00:13 new-script.sh
-rwxr-xr-x. 1 root root   0 May  7 04:09 script.sh
```

Script file:

```
#!/bin/bash

if [[ -z $1 ]] ; then
        echo "Please enter argument!!!";
        exit 1;
elif [[ $(whoami) != "root" ]] ; then
        echo "This script must run only by root";
        exit 1;
elif [[ $# -lt 3 ]] ; then
        echo "enter more than 3 arguments"
        exit 1;
else
        echo "$@"
fi
```

```
[root@rhel-9 /script #] ./script.sh
Please enter argument!!!
[root@rhel-9 /script #] su - ehmed
Last login: Thu May  7 04:12:35 +04 2026 on pts/0
[ehmed@rhel-9 ~ $] cd /script/
[ehmed@rhel-9 /script $] ./script.sh smth
This script must run only by root
[ehmed@rhel-9 /script $] su -
Password:
Last login: Thu May  7 04:13:22 +04 2026 on pts/0
[root@rhel-9 ~ #] cd /script
[root@rhel-9 /script #] ./script.sh smth1
enter more than 3 arguments
[root@rhel-9 /script #] ./script.sh smth1 smth2
enter more than 3 arguments
[root@rhel-9 /script #] ./script.sh smth1 smth2 smth3
smth1 smth2 smth3
```

Question 4

Let’s create the files:

```
[root@rhel-9 /script #] touch usernames
[root@rhel-9 /script #] touch passwords
```

Then fill them.

```
[root@rhel-9 /script #] cat usernames
Ehmed
Nihad
Mehemmed
Ravan
[root@rhel-9 /script #] cat passwords
Pass_123
Pass_1234
Pass_12345
Pass_123456
```

Then create the script file:

```
[root@rhel-9 /script #] touch create_users.sh
```

```
[root@rhel-9 /script #] chmod +x create_users.sh
[root@rhel-9 /script #] ll | grep "create_users.sh"
-rwxr-xr-x. 1 root root   0 May  7 18:35 create_users.sh
```

The script:

```bash
#!/bin/bash

usernames=($(cat /script/usernames));
passwords=($(cat /script/passwords));
for i in 0 1 2 3; do
        username=${usernames[$i]};
        password=${passwords[$i]};

        if grep -q "$username" /etc/passwd ; then
                echo "User ${username} already exists";
                continue;
        fi

        useradd -m "$username";
        echo "$password" | passwd --stdin "$username";
        echo "User ${username} has created!";
done

mkdir -p /backups

for username in "${usernames[@]}"; do
    tar -czf /backups/${username}_home.tar.gz -C /home "$username"
    echo "Backup created for $username"
done
```

```
[root@rhel-9 /script #] ls /home
Nihad  Ravan
[root@rhel-9 /script #] ./create_users.sh
Changing password for user Ehmed.
passwd: all authentication tokens updated successfully.
User Ehmed has created!
User Nihad already exists
Changing password for user Mehemmed.
passwd: all authentication tokens updated successfully.
User Mehemmed has created!
User Ravan already exists
Backup created for Ehmed
Backup created for Nihad
Backup created for Mehemmed
Backup created for Ravan
[root@rhel-9 /script #] ls /home
Ehmed  Mehemmed  Nihad  Ravan
[root@rhel-9 /script #] ls /backups
Ehmed_home.tar.gz  Mehemmed_home.tar.gz  Nihad_home.tar.gz  Ravan_home.tar.gz
```

Question 5

Let’s download the package to send mail:

```
[root@rhel-9 /script #] dnf whatprovides mailx
Last metadata expiration check: 0:21:13 ago on Thu 07 May 2026 07:12:34 PM +04.
s-nail-14.9.22-6.el9.x86_64 : Environment for sending and receiving mail
Repo         : AppStream-Repo
Matched from:
Provide      : /bin/mailx
Filename     : /usr/bin/mailx

[root@rhel-9 /script #] dnf install s-nail-14.9.22-6.el9.x86_64 -y
Last metadata expiration check: 0:21:27 ago on Thu 07 May 2026 07:12:34 PM +04.
Dependencies resolved.
================================================================================
 Package                          Architecture                     Version
================================================================================
Installing:
 s-nail                           x86_64                           14.9.22-6.el9
```

Let's create the file:

```
[root@rhel-9 /script #] touch disk_check.sh
[root@rhel-9 /script #] chmod +x disk_check.sh
[root@rhel-9 /script #] ll | grep "disk_check"
-rwxr-xr-x. 1 root root   0 May  7 19:36 disk_check.sh
```

The script:

```
[root@rhel-9 /script #] cat disk_check.sh
#!/bin/bash

df -h | awk 'NR>1 {print $5, $6}' | while read use partition
do
    percent=$(echo $use | sed 's/%//')

    if [ "$percent" -gt 80 ]
    then
        echo "$partition is above 80% usage"
    fi
done
```

- Here, `df -h` helps us to see disk spaces. `awk` command helps us to select specific one or more columns. I have selected the fifth and sixth columns which are "Use" "Monted On". Here, `NR>1` means to ignore the first line (header part).
- We give "use" name to fifth column and "partition" name to sixth column.
- `sed` command is used to replace the part of the string. Here, from "Use" column we remove "%" (replace it with nothing). So, if it is 90%, it is converted to 90.
- `-gt` means "greater than".

If we run the command:

```
[root@rhel-9 /script #] ./disk_check.sh
/mnt is above 80% usage
```

If we really check it with running `df -h`:

```
[root@rhel-9 /script #] df -h
Filesystem             Size  Used Avail Use% Mounted on
devtmpfs               4.0M     0  4.0M   0% /dev
tmpfs                  838M     0  838M   0% /dev/shm
tmpfs                  335M  1.2M  334M   1% /run
efivarfs               256K   57K  195K  23% /sys/firmware/efi/efivars
/dev/mapper/rhel-root  8.4G  4.9G  3.5G  59% /
/dev/nvme0n1p2         960M  242M  719M  26% /boot
/dev/sr0                13G   13G     0 100% /mnt
/dev/nvme0n1p1         599M  7.4M  592M   2% /boot/efi
tmpfs                  1.0M     0  1.0M   0% /run/stratisd/ns_mounts
tmpfs                  168M   36K  168M   1% /run/user/0
```

Question 6

I have created a script file:

```
[root@rhel-9 /script #] touch del-user.sh
[root@rhel-9 /script #] chmod +x del-user.sh
[root@rhel-9 /script #] ll | grep "del-user"
-rwxr-xr-x. 1 root root   0 May  7 21:25 del-user.sh
```

In that script, we can easily convert `if-elif-else` logic to `case` system:

```
case "$cavab" in
    Yes)
        userdel -r "$i"
        echo "$i istifadecisini sildiniz"
        ;;

    No)
        echo "$i useri silmediniz"
        ;;

    *)
        echo "Yes/No secmeli idiniz"
        ;;
esac
```

```
[root@rhel-9 /script #] ./del-user.sh
u11 istifadecisi elave olundu
u22 istifadecisi elave olundu
u33 istifadecisi elave olundu
u44 istifadecisi elave olundu
u55 istifadecisi elave olundu
[root@rhel-9 /script #] ./del-user.sh
u11 istifadecisi vardir
Silmek isteyirsinizmi? Yes/No
Yes
Sizin cavabiniz Yes oldu
u11 istifadecisini sildiniz
u22 istifadecisi vardir
Silmek isteyirsinizmi? Yes/No
Yes
Sizin cavabiniz Yes oldu
u22 istifadecisini sildiniz
u33 istifadecisi vardir
Silmek isteyirsinizmi? Yes/No
Yes
Sizin cavabiniz Yes oldu
u33 istifadecisini sildiniz
u44 istifadecisi vardir
Silmek isteyirsinizmi? Yes/No
Yes
Sizin cavabiniz Yes oldu
u44 istifadecisini sildiniz
u55 istifadecisi vardir
Silmek isteyirsinizmi? Yes/No
Yes
Sizin cavabiniz Yes oldu
u55 istifadecisini sildiniz
```